

Oliver Bilston

✉ bilstonoliver@gmail.com ☎ 0426 562 908 in Oliver Bilston 🌐 Portfolio Website

Profile

Mechanical Engineering and Business Analytics student with hands-on experience across professional aviation, mechanical, electromagnetic, and naval design. Proven communicator and collaborator - having led a 150+ person engineering team as COO while simultaneously delivering in fast-paced, professional aerospace environments. Comfortable managing competing priorities across multidisciplinary teams and adapting quickly when it counts. A fast learner with a passion for problem-solving, who is highly self-motivated and creative. Loves the outdoors, winter sports, music and travelling.

Education

Monash University <i>Bachelor of Mechanical Engineering (Honours) and Business Analytics</i>	<i>Jan 2022 – Current</i>
KU Leuven - Exchange Program <i>Bachelor of Mechanical Engineering (Honours) and Business Analytics</i>	<i>Feb 2025 – Jun 2025</i>

Technical Experience

Undergraduate Design Engineer <i>Five Rings Aerospace</i>	<i>Nov 2025 – Current</i> <i>Tullamarine, VIC</i>
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- Designed 50+ airworthiness modifications for civil and specialised aircraft categories, including but not limited to - temperature regulation systems, structural modifications (Door panels, secondary structure modification), crack and corrosion repairs, livery installations, equipment removals and more.
- Ensured modifications met CASA, EASA and FAA regulatory standards
- Employed engineering principles. Load path analysis, Fastener strength analysis, Fatigue Analysis, Electrical load analysis - utilising analytical and finite element methods to ensure my designs were safe.
- Engaged with clients, ensuring project scope was consistent and they stayed informed and happy with the status of the work
- Engineered and deployed multiple custom automation tools that eliminated high-volume manual workflows, reducing project turnaround processes from months to seconds and directly improving quote accuracy and operational output

Chief Operating Officer <i>Monash Uncrewed Aerial Systems</i>	<i>Sep 2025 – Current</i> <i>Clayton, VIC</i>
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- Manages 130+ engineers across Aerostructures, Avionics, Flight Operations, Propulsion and Operations.
- Directly engaged with key stakeholders, including university faculty, government bodies, financial and in-kind sponsors, including CubePilot, Altium, and Leap Australia
- Coordinating development of 1 multirotor and 2 fixed-wing humanitarian uncrewed aircraft.
- Coordinating international travel to attend the SUAS competition in Tulsa, OK, September 2026.
- Overhauled systems engineering processes and documentation frameworks from the ground up, implemented industry-style practices that significantly improved team efficiency and project delivery quality

Motor Development Lead <i>Monash Uncrewed Aerial Systems</i>	<i>Jan 2024 – Sep 2025</i> <i>Clayton, VIC</i>
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- Designed a fully custom brushless DC motor from first principles using advanced electromagnetic, structural, and thermal FEA simulation, achieving a 150g weight reduction and 15% efficiency improvement over off-the-shelf solutions

Research Project - Development of a autonomous multimodal capable, Sail Powered ASV for low impact shallow reef and ocean surveying <i>Monash University</i>	<i>Clayton, VIC</i> <i>July 2025 – Current</i>
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- Conducted naval architecture and aerodynamic analysis to design a fully autonomous sailboat, validating hydrodynamic and sail performance through simulation as well as physical testing
- Fabricated structural components hands-on across composite layup, metalwork, and surface finishing
- Coordinated system integration across a multidisciplinary team, managing timelines and contributing to electrical decision-making alongside other engineers

Medical Administrative Assistant

Skin Cancer Day Surgery

Ashwood, VIC

Apr 2024 – Nov 2025

- Organised surgeries and consultations for elderly patients with dermatologists.
- Coded medical procedures and created Workplace Instructions (WI's) to improve workplace efficiency and training efficacy

Ski Lift Operator

Sunshine Village Ski Resort

Alberta, Canada

Nov 2023 – Mar 2024

- Operated heavy machinery and ensured customer safety on ski lifts.
- Worked in extreme conditions, engaged customers and accomodating needs
- Worked in conjunction with maintenance teams to reliably run multimillion dollar ski lifts.

Senior Bartender and Waiter

La Cabra Mexican

Camberwell, VIC

Jun 2022 – Nov 2023

- Sole bartender for a very busy restaurant, juggling stock mangement, customer orders and organisation
- Also waited tables, engaging with customers and ensuring a positive experience
- Ensured efficiency and organisation in a high pressure working environment

Night Supervisor and Front End Staff

Woolworths

Hawthorn East, VIC

Jun 2020 – Jun 2022

- Nighttime supervisor for a large Woolworths, managed tills, cleanliness and basic stock levels
- Skilled at customer engagement and resolving issues

Skills

Languages: Matlab, Python, JavaScript, SQL, C++

Technologies: SolidWorks, Ansys, Creo, Simulink, Blender, Photoshop, Microsoft 365, Fusion 360, Autocad

Fabrication: Additive Manufacturing, Light Fabrication, Metalworking, Soldering, Rapid Prototyping

Design Skills: Structural Analysis and Design, Finite Element analysis validation, Design for Manufacturing principles, Electromagnetic Design, Endurance and Power system design, Fluid Dynamics, Thermodynamics, Vibrationary Analysis

Soft Skills: Excellent Manager and Communicator, Easy Collaborator, Well documented, Good Time Management, Team Player

Accreditation

Certificates: Finite Element Analysis (FEA) and Computational Fluid Dynamics (CFD) (Leap Australia), ASSOCIATE - Mechanical Design (Dassault Systems), Light Fabrication, Composites, Low Voltage, Responsible Service Of Alcohol, Food Safety, Silver Service

Hobbies

- Skiing
- Hiking
- Guitar
- Travelling